

CU Online — Chandigarh University

MCA Final Year Project — Milestone Report

MILESTONE 3 REPORT

Store Cost Comparison Data

Designing the data schema, connecting Power BI, and applying Power Query transformations

Project Title: Multi-Cloud Cost Comparison and Optimization Analytics Dashboard

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Milestone 3: Store Cost Comparison Data

milestone focused on the data storage and transformation layer of the system. With the simulation dataset from Milestone 2 complete, this milestone addressed how that data would be structured for analytical use — specifically, the relational schema design, the Power BI connection to the Excel source, and the Power Query transformation pipeline that cleans and normalises the raw data before it reaches the analytical layer.

3.1 Data Schema Design

The schema was designed around five related tables, each serving a distinct analytical purpose. The primary design principle was to separate raw transactional data from derived aggregates, placing pre-computed aggregates in the Annual Summary and Quarterly Breakdown sheets to reduce the computational load on Power BI's DAX engine at query time.

The Raw Dataset table uses a composite primary key of (Financial Year, Quarter, Provider) — a three-field combination that uniquely identifies each of the 60 records in the table. This composite key was chosen over a surrogate integer primary key because the natural key is meaningful to users and aligns with how analysts think about the data (they will naturally think in terms of 'FY2023 Q2 GCP costs', not 'record 34').

3.2 Power BI Data Connection Established

Power BI Desktop was connected to the Excel source file using the Get Data > Excel Workbook connector. The Navigator showed all five sheets. All five were selected for import into the data model. The Transform Data option was selected (rather than Load directly) to open the Power Query Editor for data cleaning.

3.3 Power Query Transformation Pipeline

The Power Query transformation pipeline was designed and implemented during this milestone. The pipeline consists of five applied steps for the Raw Dataset sheet (the most complex transformation):

1. Source — Excel.Workbook connection to the XLSX file with header promotion enabled.
2. Navigation — Selection of the 'Raw Dataset' sheet from the workbook content list.
3. PromoteHeaders — Promotion of the first row to column headers using Table.PromoteHeaders.
4. RemoveLastRow — Removal of the TOTAL/AVG summary row using Table.RemoveLastN(table, 1). This step was identified as necessary after initial testing revealed that the summary row was being included in Power BI aggregations, inflating KPI card values and generating spurious data points in bar charts (BUG-002, resolved here).
5. ChangedTypes — Assignment of correct data types to all nine columns using Table.TransformColumnTypes. This step was added after discovering that the Cost (\$M) column was being imported as text type due to floating-point binary representation artifacts (BUG-001, resolved here).

3.4 Bug Identification and Resolution

Bug ID	Symptom	Root Cause	Resolution	Status
BUG-001	Cost (\$M) displays as 9.800000000000001	IEEE 754 floating-point binary representation of decimal 9.8 imported as text	Explicit Table.TransformColumnTypes assigning type number to Cost (\$M) column; FORMAT display measure added to KPI card	Resolved
BUG-002	TOTAL/AVG summary row appearing as data point in bar chart	Excel summary row imported as data record	Table.RemoveLastN(table, 1) step added to pipeline	Resolved

Table M3.1: Bugs Identified and Resolved in Milestone 3

3.5 Data Model Relationship Definition

After completing the Power Query transformations and loading the cleaned data, the table relationships were reviewed in Power BI's Model view. The five tables are connected through the Provider field (a common dimension across all transactional tables). The Raw Dataset and Quarterly Breakdown tables share the same composite key structure and can be connected through the Provider dimension for visual cross-filtering. The Free Tier and Recommendations tables are standalone reference tables with no formal model relationships, bound directly to their respective visuals through field selection.

3.6 Milestone Deliverables Completed

- Five-sheet Excel workbook schema finalised and documented.
- Power BI data connection established to all five sheets.
- Power Query transformation pipeline implemented and tested for the Raw Dataset sheet.
- Simplified transformation (Source, Navigate, PromoteHeaders, ChangedTypes) applied to the remaining four sheets.
- BUG-001 (floating-point display) and BUG-002 (summary row in chart) identified and resolved.
- Data model relationships defined in Power BI's Model view.